

QNT 402 Quantum Information
Homework 7

1. [35] *A simple model test.* In this problem we'll generate some fake data and then try to fit it to determine if modeling it as linear or quadratic function is more reasonable.

- a.) [5] First generate some fake data using the Python code:

```
slope = 1
offset = 0.25
time = np.linspace(0,10,10)
data = np.random.normal(slope*time + offset,1)
sigma = 1*np.random.normal(np.ones(len(data)),0.01)
```

and plot it with the pyplot errorbar() function. This is obviously linear data but it's pretty noisy as you'll see.

- b.) [15] Now use curve_fit to fit a linear function ($bx + a$) and a quadratic function ($cx^2 + bx + a$) and add the plot of the best fits to the plot with the data on it.
- c.) [15] Now calculate the χ^2/ν of each of these and try to determine which is a more likely model.

2. [30] *Fisher Information.*

- a.) [15] In class we found that using the typical phase sensing protocol with N_q qubits in a GHZ state produced the state $|\psi\rangle = \sin \frac{N_q \phi}{2} |0\rangle + \cos \frac{N_q \phi}{2} |1\rangle$. We showed that this would allow sensing of the phase with precision: $\sigma_\phi = 1/(N_q \sqrt{N})$, where N is the number of repetitions of the experiment. Calculate the Cramer-R  o bound, $\sigma_\phi^{(CR)}$, for the phase sensitivity and compare it to σ_ϕ .

- b.) [15] Suppose we are trying to measure the oscillator phase ϕ using a coherent state $|\alpha\rangle$ by measuring the position x . Writing $\alpha = |\alpha|e^{i\phi}$, the wavefunction of a coherent state is

$$\psi(x, |\alpha|, \phi, t) = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}(x - \langle x(|\alpha|, \phi) \rangle)^2 + \frac{i}{\hbar} \langle p(|\alpha|, \phi) \rangle x - \frac{i}{2}(\omega t - |\alpha|^2 \sin(2\omega t - \phi))}$$

where $\langle x(|\alpha|, \phi) \rangle$ and $\langle p(|\alpha|, \phi) \rangle$ are the expectation value of the position and moment of the coherent state and I've made their dependence on $|\alpha|$ and ϕ explicit. Calculate the Cramer-R  o bound for the measurement of ϕ .

3. [50] *Alright, alright, alright, let's work with some quantum data.* In this problem we will generate data from a quantum sensing experiment. We know that for a Ramsey sequence (we worked it out last quarter) that for a qubit starting in $|0\rangle$ the probability of finding the qubit in state 1 is given as:

$$P_1 = 1 - \frac{\Omega^2}{\Omega'^4} \left(\Omega' \cos\left(\frac{\delta t_w}{2}\right) \sin(\Omega' t_p) - 2\delta \sin\left(\frac{\delta t_w}{2}\right) \sin^2\left(\frac{\Omega' t_p}{2}\right) \right)^2$$

This expression is for two $\pi/2$ pulses of length t_p separated by a free evolution time of t_w .

- a.) [20] Write a Python function that generates realistic data for N repetitions of such a sequence. This function will output a value for P_1 that comes from a binomial distribution described by the true value of P_1 and the number of trials N . Use this

function to make a plot of the “measured” values of P_1 versus t_w for the parameters $\Omega = 2\pi$ and $\delta = \frac{\pi}{4}$ for $N = 1, 10, 100, 1000$.

- b.) [20] Now simulate the signal for what is called “slope detection”. Find the wait time for the first $P_1 = \frac{1}{2}$ point (i.e. the one near $t_w = 2$) and plot P_1 vs Φ for $-1 \leq \Phi \leq 1$. Assume $\gamma = 1$. Make the plots for $N = 1, 10, 100, 1000$. Compare your result to that derived in Section IV.E.1 in of Reviews of Modern Physics, Volume 89, 035002 (2017)
 - c.) [10] Take the plot you made for $N = 100$ and add the expected signal plot to it as well as the expected signal plus σ_p and the expected signal minus σ_p . Is it what you expect?
4. [60] *One last dance with the QHO*. Here, we’re going to examine the effect of squeezing on a QHO.
- a.) [10] Define a Python function that takes time, t , as its input and returns the time evolution operator a time t for the Hamiltonian $H_o = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right)$. Use that function to calculate $|\alpha(t)\rangle = \hat{U}(t)|\alpha\rangle$ for various times during the oscillation period of the oscillator. Plot the expectation value of position and momentum in the (x, p) plane for $\alpha = 1$ – make sure to make your basis is big enough for this size coherent state. Choose $\omega = 2\pi$ and $m = 1$. Here, you’ll likely find QuTip’s `displace()` function handy.
 - b.) [10] Calculate σ_x and σ_p at each of the times you chose in part (a) and plot σ_x , σ_p , and $\sigma_x \sigma_p$ versus time.
 - c.) [15] QuTip has a function called `squeeze()` that implements the squeeze operator we discussed in class. Create a squeezed coherent state by first applying the squeeze operator (with squeeze parameter $r = 1$; fyi, QuTip uses z for the parameter r from class) and then applying the displacement operator used in part (a). Now, remake the plots of part (a) and part(b). What do you notice?
 - d.) [20] The squeeze operator is a bit of shortcut, so let’s make sure we know how to implement it on real hardware. Add a drive to QHO of the form $V = \beta_1 x \sin(2\omega t) + \beta_2 x^2 \sin(2\omega t)$ use e.g. `sesolve` to find the evolution. Choose $\beta_1 = \beta_2 = 1$. Make the same plots you made in part (a) and part (b). Is this squeezing?
 - e.) [5] What are the two terms in the potential V ? Try setting e.g. $\beta_1 = 0$ and describe what you see.